

SYNOPSIS



At 1840 on 12 January 2011, the 16.84m prawn trawler *Silver Chord* (**Figure 1**) collided with the 14.99m prawn trawler *Sapphire II* (**Figure 2**). At the time of the collision, *Sapphire II* was stopped in the water while her skipper, who was operating single-handed, recovered the vessel's fishing gear. *Silver Chord* was making 5.5 knots (kts) as she proceeded towards Stornoway after a day's fishing.

Sapphire II's hull was penetrated and the fish hold quickly started to flood. Her skipper reported the accident to the coastguard, and transferred to *Silver Chord* before *Sapphire II* sank at about 1850. *Silver Chord* was also damaged, but was able to return to Stornoway under her own power. There were no injuries.

The investigation identified a number of factors which contributed to the collision. These included:

- Neither skipper was aware of the proximity of the other vessel despite both vessels displaying navigation lights and deck floodlights, and both vessels being equipped with working radar displays.
- *Silver Chord*'s wheelhouse was left unattended for substantial periods of time before the accident while her skipper and crew worked in a fully enclosed shelter.
- The visibility ahead from *Silver Chord*'s wheelhouse was impaired by the vessel's shelter and deck fittings.
- *Sapphire II*'s skipper had been focused on recovering the fishing gear. His view from the aft working deck was obstructed by the vessel's accommodation and wheelhouse, which prevented him from maintaining a proper and effective lookout. The single-handed operation of *Sapphire II* was therefore inherently unsafe.

This is one of many accidents involving fishing vessels in which leaving the wheelhouse unattended has been a significant contributory factor. Despite Maritime and Coastguard Agency (MCA) and industry initiatives, this behaviour persists among many fishermen as economic and social factors continue to take preference over safety considerations. As leaving a wheelhouse unattended is unavoidable during single-handed operations, the suitability of vessels to be operated in this way must be carefully assessed.

Recommendations have been made to the MCA aimed at ensuring that all vessels are sufficiently manned to ensure their safe operation, and that the current requirements for the visibility from fishing vessels' wheelhouses are met. Recommendations have also been made to the owner/skippers of *Silver Chord* and *Sapphire II* which are intended to improve the safe operation of their vessels in the future.

1.2 NARRATIVE

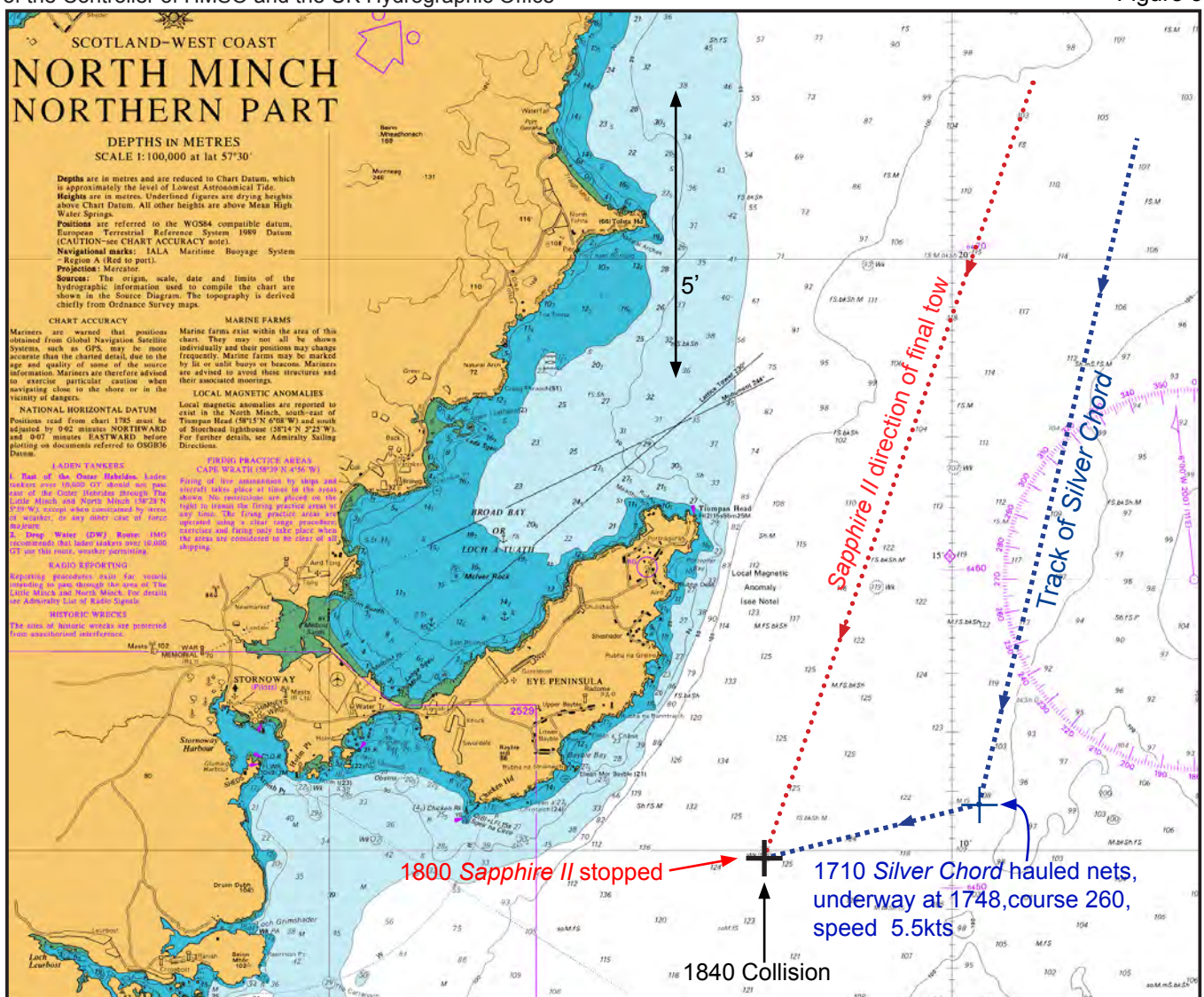
1.2.1 *Sapphire II*

Shortly after 0600 on 12 January 2011, *Sapphire II* (**Figure 2**) sailed from the fish quay in Stornoway with her owner/skipper on board. The skipper shot away the vessel's nets at 0820 and started his first tow of the day. The tow was completed at 1300.

At 1325, *Sapphire II* was turned towards the south-south-west and the skipper shot away the nets for a second tow. At about 1800, when the vessel was approximately 3 miles off the coast (**Figure 3**), the skipper adjusted the vessel's heading to about 140° to put the wind on the starboard quarter in preparation for recovering the fishing gear. He then put the engine into neutral and switched on the fore deck floodlights. *Sapphire II*'s navigation lights and lights to indicate that she was engaged in trawling were already on. The skipper could see the lights of other vessels on the horizon to the north-east but he was unable to identify them.

Reproduced from Admiralty Chart BA 1785 by permission of the Controller of HMSO and the UK Hydrographic Office

Figure 3



Extract of Chart BA 1785 showing the tracks of *Silver Chord* and *Sapphire II*

The skipper left the wheelhouse to go to the working deck. He recovered 400m of trawl wire, unclipped the trawl doors, brought the spreaders inboard, moved the net forward over the bag hatch, and released the catch into the hopper. During this period, the skipper frequently moved between the winch controls under the whaleback (**Figure 2**) and the stern. By about 1830, the skipper had recovered the net, and was standing on the starboard quarter of the aft deck ready to bring the dog rope inboard when, at 1840, he felt an impact and heard a massive thud forward.

1.2.2 *Silver Chord*

At 0630 on 12 January 2011, the prawn trawler *Silver Chord* (**Figure 1**) sailed from Stornoway for a day's fishing. On board were her owner/skipper and two deckhands. The first tow took place between 0830 and 1230; during this time, the skipper took the opportunity to get some rest while the two deckhands took turns keeping a watch in the wheelhouse.

A second tow was conducted between 1315 and 1700, following which, the skipper stopped *Silver Chord* in the water and the net was hauled. The process took longer than usual because the net was tangled. However, by 1748 the net had been recovered and *Silver Chord* began steaming back to Stornoway on a heading of 260° at 5.5kts (**Figure 3**). The vessel's navigation lights and her aft floodlights were on; it is not known if her all-round green and white trawling lights were also lit. The skipper intended to pass 1 mile to the south of Chicken Rock, which was his preferred route into Stornoway when approaching from the east.

Once *Silver Chord* began making way, the skipper engaged the autopilot. As he did so, he scanned the horizon ahead and noticed another fishing vessel, which he recognised as *Kaylana*, between 2 and 3 miles off the starboard bow, which was also heading back to Stornoway. The radar display was set to the 3-mile range scale and the only target seen by the skipper was that of *Kaylana*.

The skipper then left the wheelhouse to help the two deckhands sort the catch in the fully enclosed shelter. He returned to the wheelhouse on four or five occasions to check on the whereabouts of *Kaylana* but he did not see any other vessels or targets on the radar display.

At approximately 1840, a heavy impact was felt by the skipper and the deckhands while working under the shelter, and *Silver Chord* came to a sudden halt.

1.3 ABANDONMENT, RESCUE AND DAMAGE

After hearing the thud caused by the collision, *Sapphire II*'s skipper immediately ran forward from the starboard quarter of the aft deck, and saw *Silver Chord* embedded in the port bow at the waterline. He immediately opened the fish hold hatch and saw that this compartment was flooding rapidly. The skipper was concerned that the bilge pumps would not cope with the rate of water ingress. He rushed to the wheelhouse and, at 1842, he informed Stornoway coastguard on very high frequency (VHF) radio channel 16 that *Sapphire II* had been in a collision and was taking in a lot of water. The coastguard immediately tasked a search and rescue (SAR) helicopter and the Stornoway-based lifeboat to assist.

Meanwhile, *Silver Chord's* skipper ran to the wheelhouse and saw the bow of his vessel embedded into the port bow of *Sapphire II*. The skipper was shocked, but he immediately put the engine control to astern until *Silver Chord's* stem came clear. He then searched for damage; the two deckhands continued to sort the catch inside the shelter.

On returning to the deck, *Sapphire II's* skipper saw that the fish hold was now almost half-filled with water and that *Silver Chord* had manoeuvred clear. He tried to contact *Silver Chord* on VHF channel 6 but there was no response. He then attracted the attention of *Silver Chord's* skipper by shouting a request for *Silver Chord* to manoeuvre towards *Sapphire II's* stern and be ready to take him off. This was heard by *Silver Chord's* skipper, who manoeuvred his vessel accordingly.

As *Silver Chord* closed, the skipper of *Sapphire II* felt his vessel's stern rising and realised that she was about to founder. He quickly picked up his mobile phone and tobacco from the wheelhouse and stepped across onto *Silver Chord's* deck.

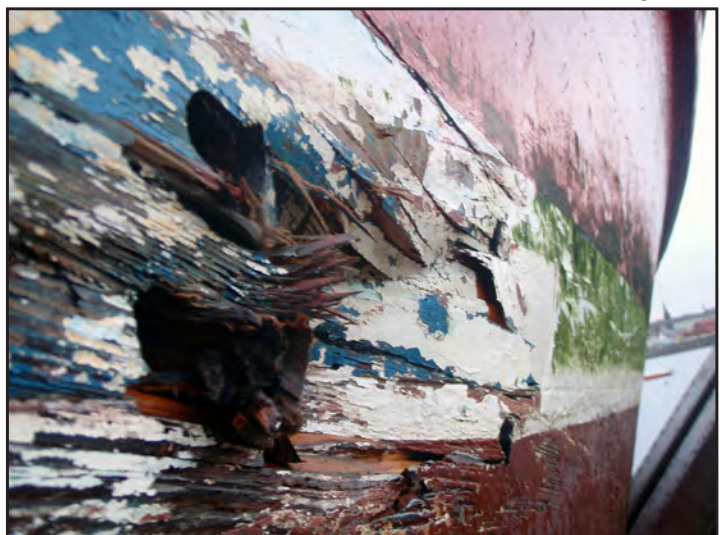
At 1850, the bow of *Sapphire II* began to submerge and rotate to port. Simultaneously, the vessel's stern rose into the air and the vessel quickly foundered. Seconds later, the vessel's inflated liferaft and her Emergency Position Indicating Radio Beacon (EPIRB) surfaced. A transmission from the EPIRB was received by Falmouth coastguard soon afterwards.

At 1851, *Sapphire II's* skipper informed Stornoway coastguard that he was on board *Silver Chord* and that *Sapphire II* had foundered. The coastguard instructed *Silver Chord's* skipper to stay on scene until the helicopter and the lifeboat arrived. The skipper conducted further checks for damage and found water entering the fore peak. However, the water level was being contained by the vessel's pumps.

The SAR helicopter arrived on scene at 1904 followed soon after by the lifeboat. At 1922, a portable pump and an Royal National Lifeboat Institution (RNLI) crewman were transferred from the lifeboat to *Silver Chord* to assist with the efforts to control the flooding in the fore peak. The lifeboat then recovered *Sapphire II's* liferaft and EPIRB.

Silver Chord returned to Stornoway at slow speed in order to minimise the rate of water ingress forward. She berthed safely alongside in Stornoway at 2040. On her arrival, local police breathalysed both skippers, but no trace of alcohol was found.

On 14 January 2011, *Silver Chord* was taken out of the water and inspected. The hull was damaged in way of the stem bar on both port and starboard sides around the boot topping area. On the port side, two planks had been damaged (**Figure 4**) and the hull had been penetrated at the waterline in way of the fore peak space. On the starboard side there was evidence of impact damage, but no penetration of the hull.



Damage to *Silver Chord's* port bow

Figure 4

1.11.2 *Sapphire II*

The skipper of *Sapphire II* was required by the Under 15m Code to undertake health and safety risk assessments, but these did not have to be written down. The skipper had identified that the risks associated with sailing single-handed were greatly increased, and had implemented several control measures, including a mechanical catch handling system, designed to reduce the time and risks of operating single-handed on deck. However, he had not identified the hazard of leaving the wheelhouse unattended when working on deck with the vessel underway, and therefore had not put appropriate control measures in place.

1.12 PREVIOUS ACCIDENTS

1.12.1 MAIB database statistics

Tables 1 and **2** show that since 1996, 19 of the 94 fishing vessels less than 24m in length that were involved in a collision had no-one in the wheelhouse. Eight of the collisions occurred during darkness, ten during daylight and one during twilight. Between 2002 and 2010, 5 fishing vessels under 24m grounded without a watchkeeper being in the wheelhouse.

1.12.2 Previous accidents and recommendations

In 2002, the MAIB conducted a preliminary examination into a collision between two fishing vessels, *Aspire* and *Maricos*. A contributory factor to the accident was that the shelter on *Aspire* restricted the watchkeeper's visibility when looking directly ahead. Consequently, the MCA was recommended to:

'Include the problem of poor visibility from the wheelhouse in its ongoing review of survey procedures and risk assessment for fishing vessels'.⁴

The circumstances of the collision and the recommendation were discussed at a Fishing Industry Safety Group (FISG) technical sub-group meeting, but it was concluded that no further action would be taken as appropriate guidance was already in place.

In 2006, the MAIB investigated the grounding and subsequent foundering of the fishing vessel *Greenhill*⁵ during which the vessel's two deckhands died. A causal factor to the grounding was that the wheelhouse was left unattended while the skipper and deck crew worked on the catch within the confines of the shelter.

The MAIB investigation report recommended the FISG to:

'Explore ways of improving the standard of watchkeeping on fishing vessels, giving emphasis to the importance of not leaving the wheelhouse unattended when at sea'.

The recommendation resulted in the development and implementation of a Navigation Watchkeeping Course administered by Seafish, which has subsequently developed into 2 and 5 day courses leading to skippers' certificates for fishing vessels under 16.5m in length operating within and outside 20nm respectively. To date, over 1000 fishermen have attended the 2-day course and 1500 have attended the 5-day course.

⁴ MAIB Recommendation 2003/130. Dated 30/01/09.

⁵ http://www.maib.gov.uk/publications/investigation_reports/2006/greenhill.cfm

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 THE COLLISION

When *Sapphire II*'s engine was stopped at about 1800, *Silver Chord* was heading directly towards her. However, although a risk of collision now existed, neither vessel's skipper detected the other vessel visually or by radar despite good visibility; both vessels were equipped with working radars, and in addition to displaying the required navigation lights, they had both illuminated their decks with floodlights. Consequently, neither skipper was prompted into taking any avoiding action, and *Silver Chord* ran into *Sapphire II* 40 minutes later.

The damage sustained by *Sapphire II* was caused by the impact with *Silver Chord*'s stem post, which stood proud of the hull planking. *Sapphire II* foundered quickly, within 10 minutes of the collision; the flooding in her fish room, the largest compartment within the vessel, was therefore rapid and could not have been contained by the skipper alone. His decision not to enter the fish room, but to inform the coastguard and make preparations to abandon the vessel, was sensible and entirely justified given the circumstances.

The skipper of *Silver Chord* reacted quickly to the situation and was able to establish that his own vessel had not sustained significant damage. However, in the immediate aftermath of the accident, the seaworthiness of his own vessel and the extent of any damage on *Sapphire II* would have been uncertain; it would therefore have been prudent for the skipper to have utilised the crew of his vessel to help manage the consequences of the collision rather than allowing them to continue processing the catch in the shelter.

2.3 FATIGUE

The skippers of both vessels had resumed fishing on 10 January 2011 after a 3 week break over the Christmas and New Year holiday period. Both had worked for about 13 hours on each of the 2 days before the accident. Although such long hours might be considered to be tiring, both skippers returned to port overnight and had slept well at home. Therefore, it is highly unlikely that the skippers' decision-making or actions were influenced by the effects of fatigue.

2.4 LOOKOUT

2.4.1 *Silver Chord*

Silver Chord was 3.6nm from *Sapphire II* when *Sapphire II* stopped to recover her fishing gear. Although this was beyond the range of visibility required for the navigation and trawling lights displayed by *Sapphire II* (3 miles for the white masthead light and 2 miles for the other lights), the vessel's bright deck floodlights would probably have been visible from a greater distance. With a height of eye of

approximately 3.15m, the skipper's visible horizon was 3.8 miles. Therefore, it is likely that *Sapphire II* could have been seen from *Silver Chord*'s wheelhouse as she started to head back towards Stornoway.

However, because of the restricted view ahead from the wheelhouse (**Figure 8**), the skipper's view of *Sapphire II* could have been obscured by the shelter or the deck fittings, particularly during the early stages of the passage. Nevertheless, as the distance between the vessels reduced, *Sapphire II*'s lights would inevitably have become brighter and more apparent. Therefore, notwithstanding the partially obstructed view ahead, the skipper's visual lookout during his periodic visits to the wheelhouse in the 52 minutes from leaving the fishing grounds, was not effective.

Figure 8



Silver Chord - view ahead from the steering position

Similarly, as *Silver Chord*'s radar was set to the 3-mile range scale, the radar target of *Sapphire II* would also have been displayed from about 1807 onwards. This was also not seen by the skipper. As *Sapphire II* was on a steady or near steady bearing on the ship's head throughout, it is possible the radar target of *Sapphire II* might have been occasionally obscured by the radar ship's head marker. However, it is highly unlikely to have been completely obscured each time the skipper visited the wheelhouse.

SECTION 3- CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT WHICH HAVE RESULTED IN RECOMMENDATIONS

1. Neither skipper was aware of the proximity of the other vessel. [2.2]
2. *Silver Chord's* skipper's visual assessment of the navigational situation and use of radar during his periodic visits to the wheelhouse in the 52 minutes from leaving the fishing grounds, was not effective. [2.4.1]
3. The view from where *Sapphire II's* skipper was standing on the starboard quarter of the aft deck was obstructed by the vessel's accommodation and wheelhouse. [2.4.2]
4. *Silver Chord's* skipper was predominantly working within the enclosed shelter, and therefore a continuous lookout was not maintained. [2.5.1]
5. It is disappointing that attempts to improve the safety of the navigation watch on board fishing vessels, through published guidance, risk assessments and voluntary training courses, appear to have had little impact. [2.5.1]
6. As *Sapphire II's* skipper was unable to keep a proper and effective lookout while recovering the fishing gear, the single-handed operation of *Sapphire II* was inherently unsafe. [2.5.2]

3.2 OTHER SAFETY ISSUES IDENTIFIED DURING THE INVESTIGATION ALSO LEADING TO RECOMMENDATIONS

1. There are currently no regulations requiring minimum levels of safe manning to be determined and adhered to on board UK fishing vessels. [2.5.3]
2. Unless the requirements of MGN 314 (F) are adhered to by fishing vessel owners and skippers, and enforced by MCA surveyors, restricted fields of view, from within the wheelhouse, will continue to feature as causal factors in future accidents. [2.6]

3.3 SAFETY ISSUES IDENTIFIED DURING THE INVESTIGATION WHICH HAVE NOT RESULTED IN RECOMMENDATIONS BUT HAVE BEEN ADDRESSED

1. The shelter added to *Silver Chord* in the 1980s and the fitting of deck equipment onto the shelter in the 1980s, adversely affected the visibility ahead from the wheelhouse, but no mitigating measures had been adopted. [2.6]

SECTION 4– ACTIONS TAKEN

4.1 THE SKIPPER OF *SILVER CHORD*

Has significantly improved the forward facing visibility from the wheelhouse steering position by fitting a raised observation compartment on the centreline of the wheelhouse deck head.